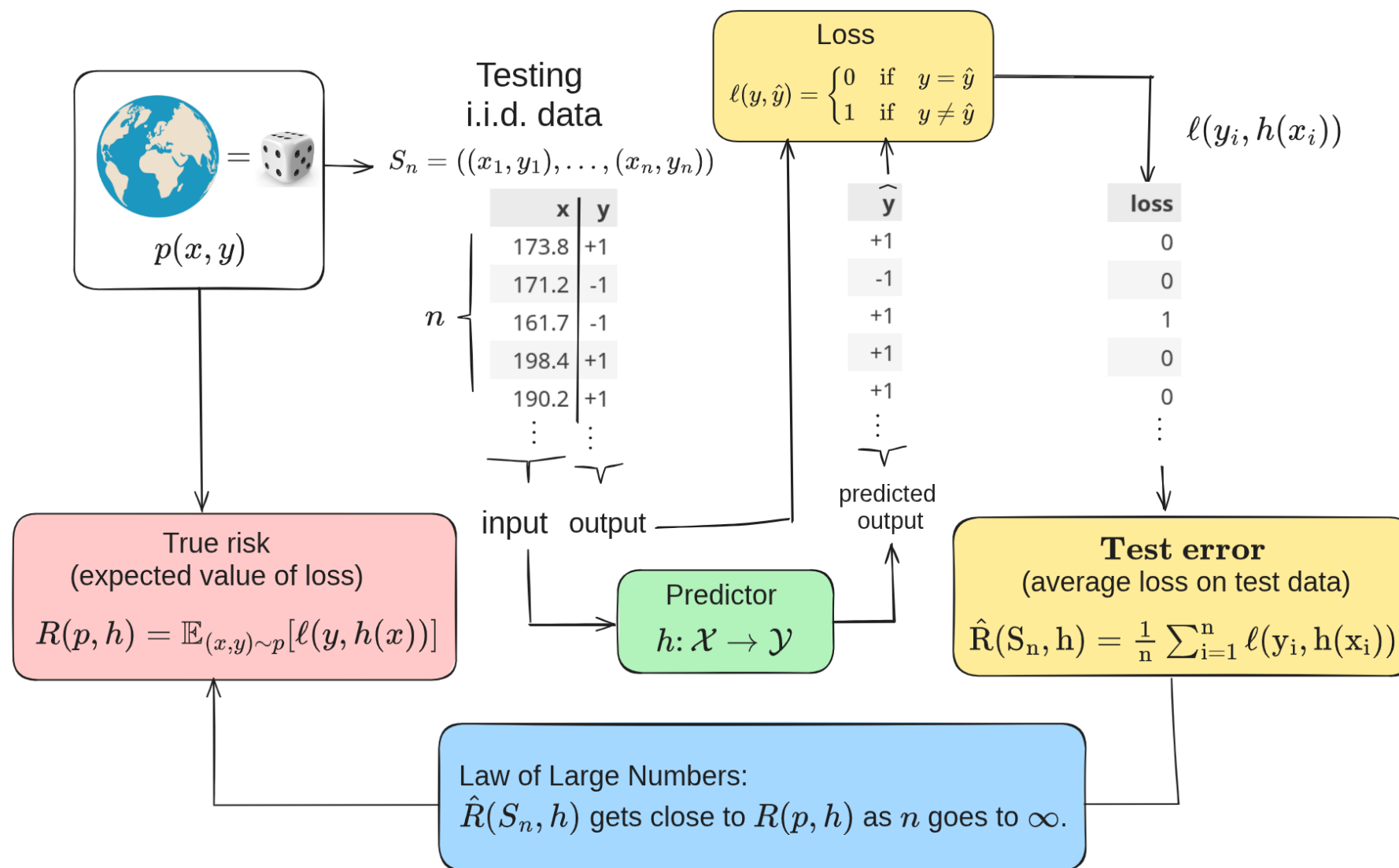


Machine Learning Fundamentals

Predictor Evaluation

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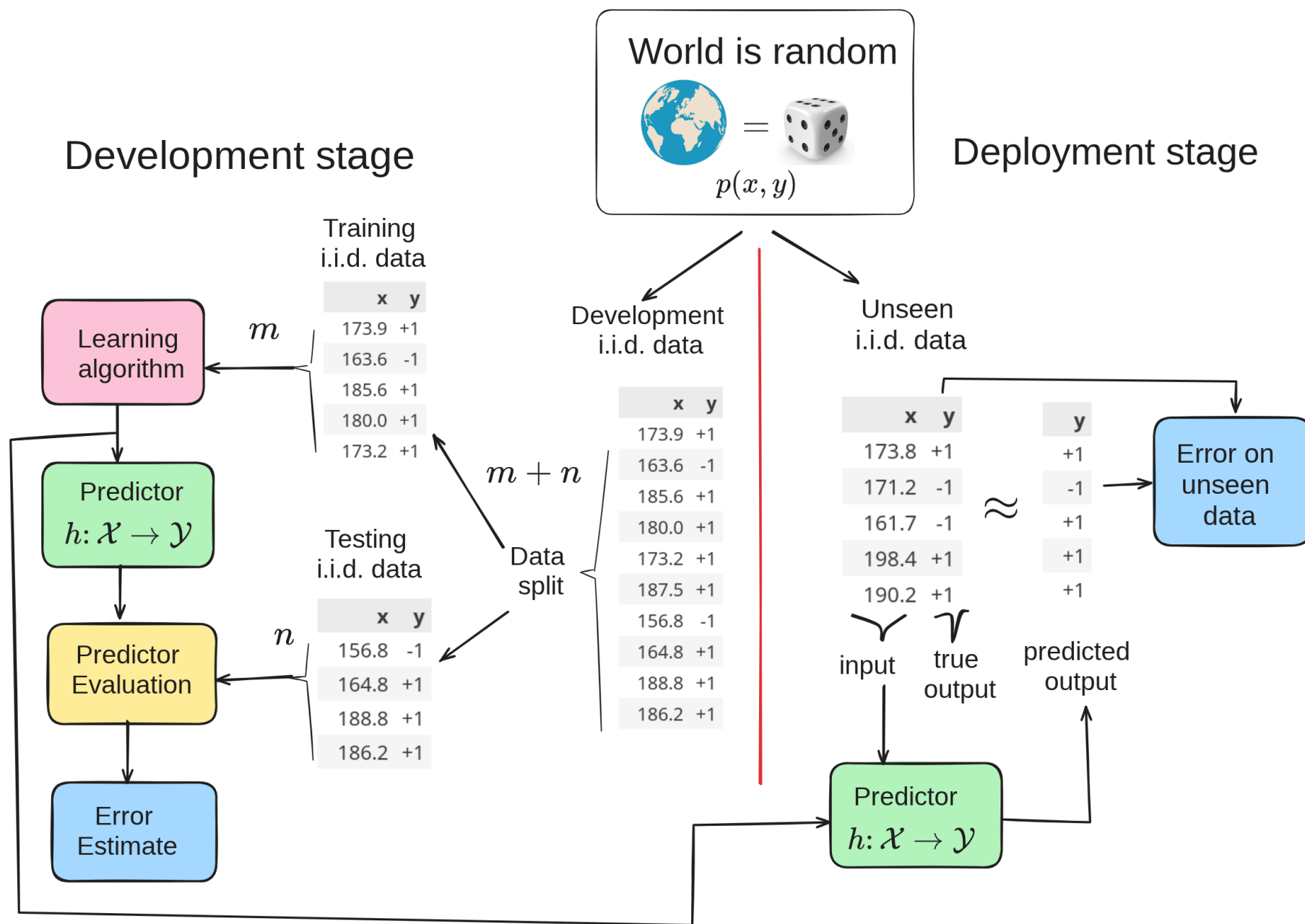
Predictor evaluation



Key observations:

- ◆ The true risk $R(h, p)$ is a fixed unknown scalar (distribution p is unknown).
- ◆ The test error $\hat{R}(S_n, h)$ is a random variable (it depends on the sample S_n), but its value can be computed once the sample is given.

Training and Testing Data Split

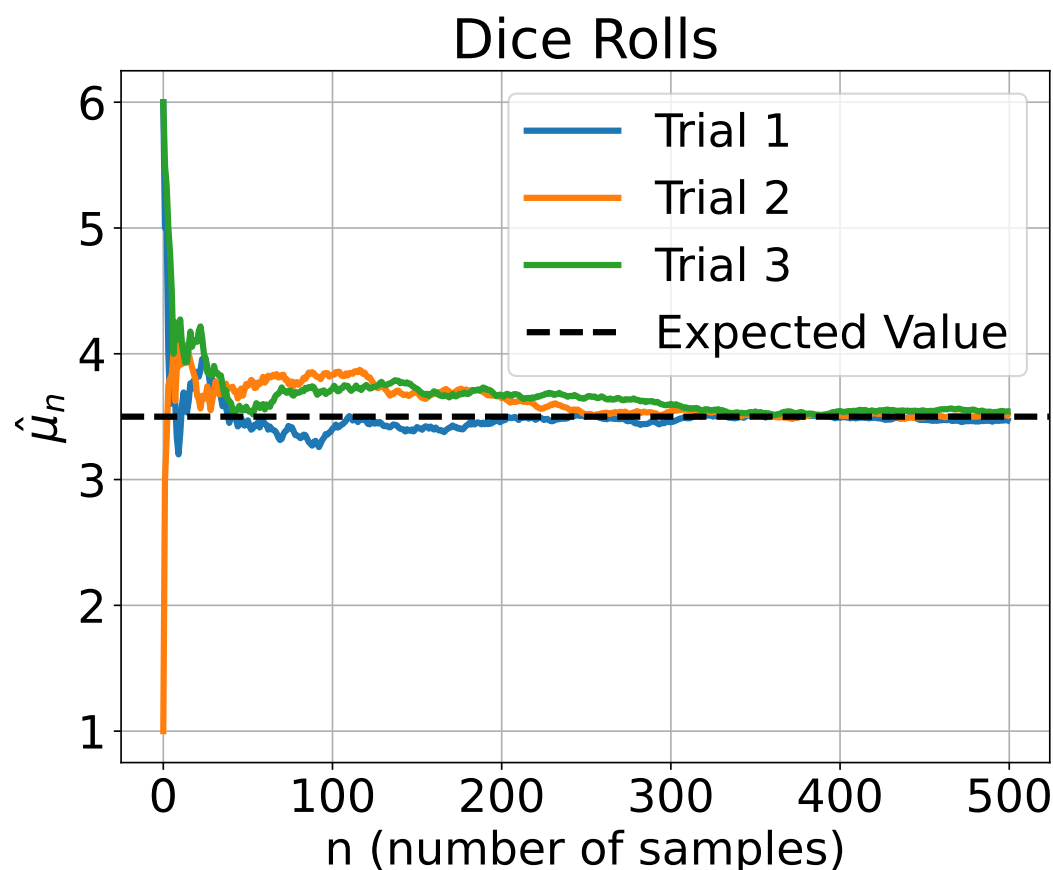
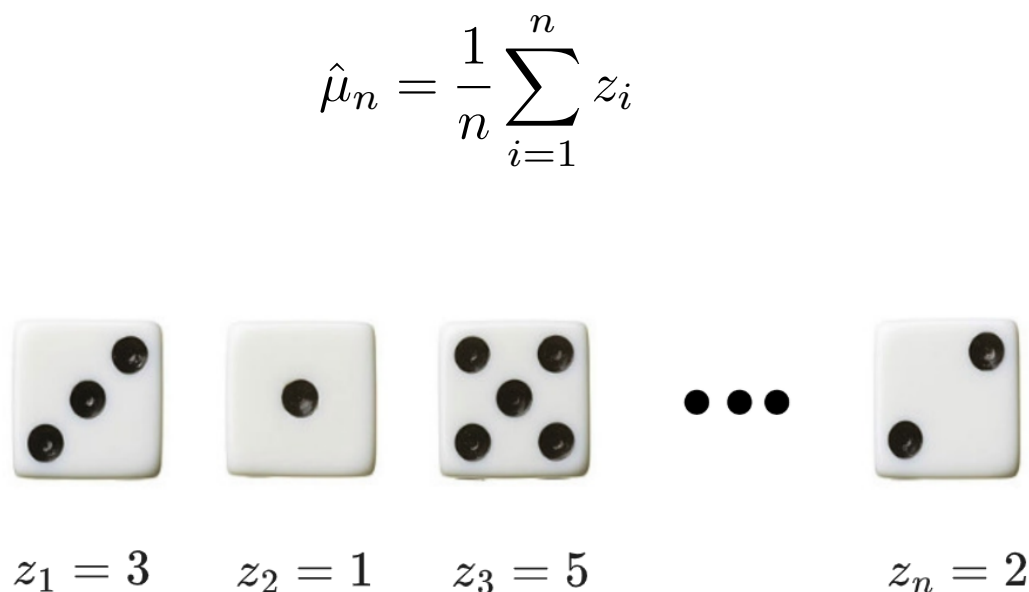


Idea: Split the available data into two disjoint parts: training data (used to learn the predictor) and testing data (used only for predictor's performance evaluation).

Law of Large Numbers

- ◆ The sample mean $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n z_i$ of i.i.d. sample $Z_n = (z_1, z_2, \dots, z_n)$ generated from $q(z)$ gets closer to the expected value $\mu = \mathbb{E}_{z \sim q}[z]$ as the sample size n increases.
- ◆ Example: The expected value of a single roll of a fair die is

$$\mu = \mathbb{E}_{z \sim q}[z] = \sum_{z=1}^6 z q(z) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$



Sample mean as an estimator of the expected value

Let $Z_n = (z_1, \dots, z_n)$ be i.i.d. sample generated from rand. vars.
with the expected value and variance:

$$\mu = \mathbb{E}_{z \sim q}[z] = \frac{1+2+3+4+5+6}{6} = 3.5$$

$$\sigma^2 = \mathbb{V}_{z \sim q}[z] = \frac{(1-3.5)^2 + \dots + (6-3.5)^2}{6} \approx 2.917$$

The sample mean

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n z_i$$

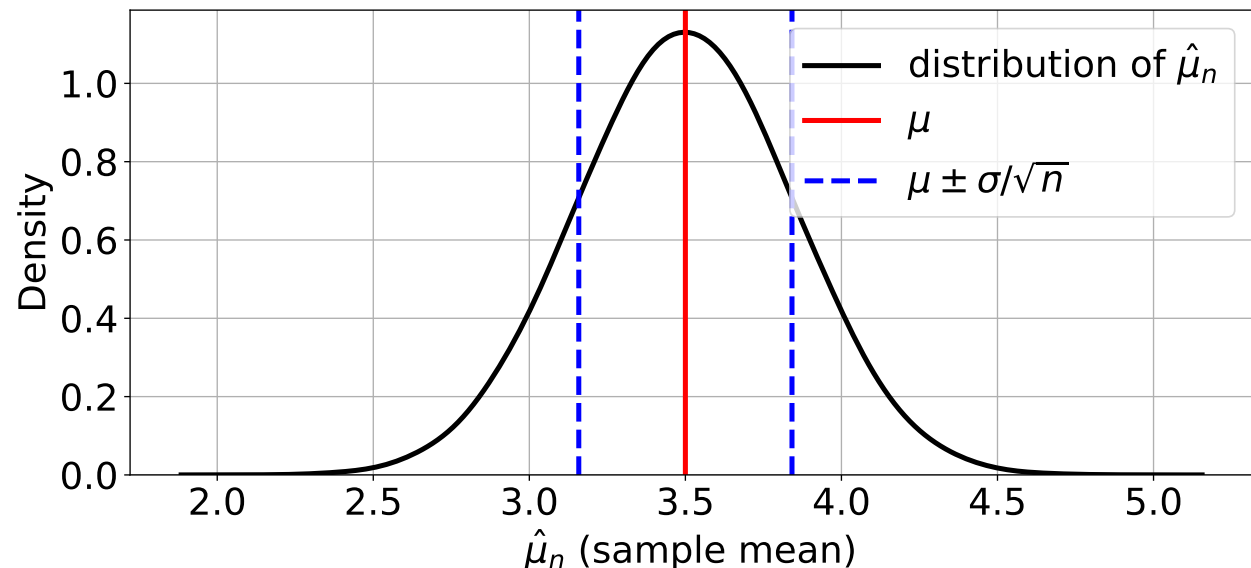
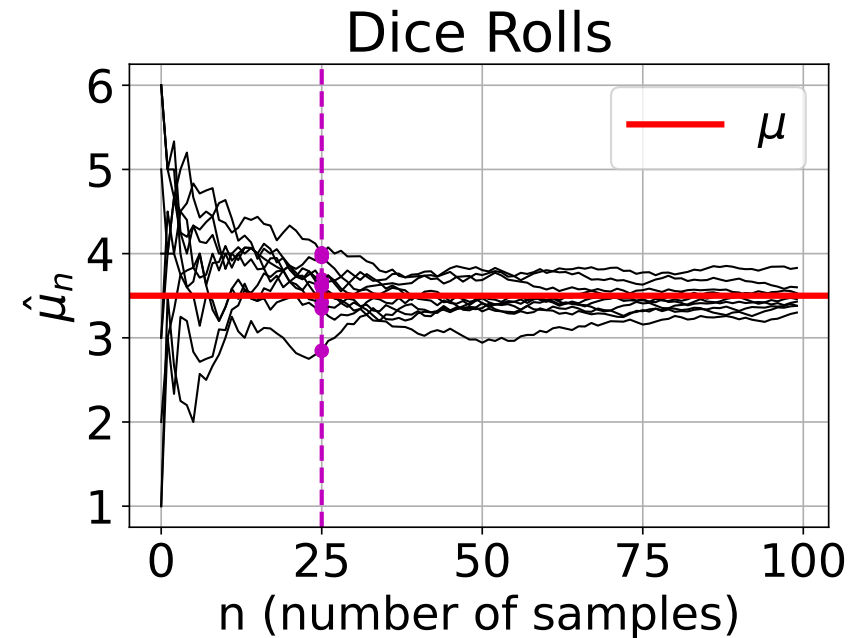
is an **unbiased estimator** of the expected value:

$$\mathbb{E}_{Z_n \sim q^n}[\hat{\mu}_n] = \mu$$

The variance of $\hat{\mu}_n$ decays with $\frac{1}{n}$:

$$\mathbb{V}_{Z_n \sim q^n}[\hat{\mu}_n] = \frac{\sigma^2}{n}$$

E.g. for die: $n = 25 \rightarrow \frac{\sigma^2}{n} = 0.116$



Sample mean as an estimator of the expected value

- ◆ Let $Z_n = (z_1, \dots, z_n)$ be i.i.d. sample generated from rand. vars. with the expected value $\mu = \mathbb{E}_{z \sim q}[z]$ and variance $\sigma^2 = \mathbb{V}_{z \sim q}[z]$.
- ◆ The sample mean $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n z_i$ is an unbiased estimator:

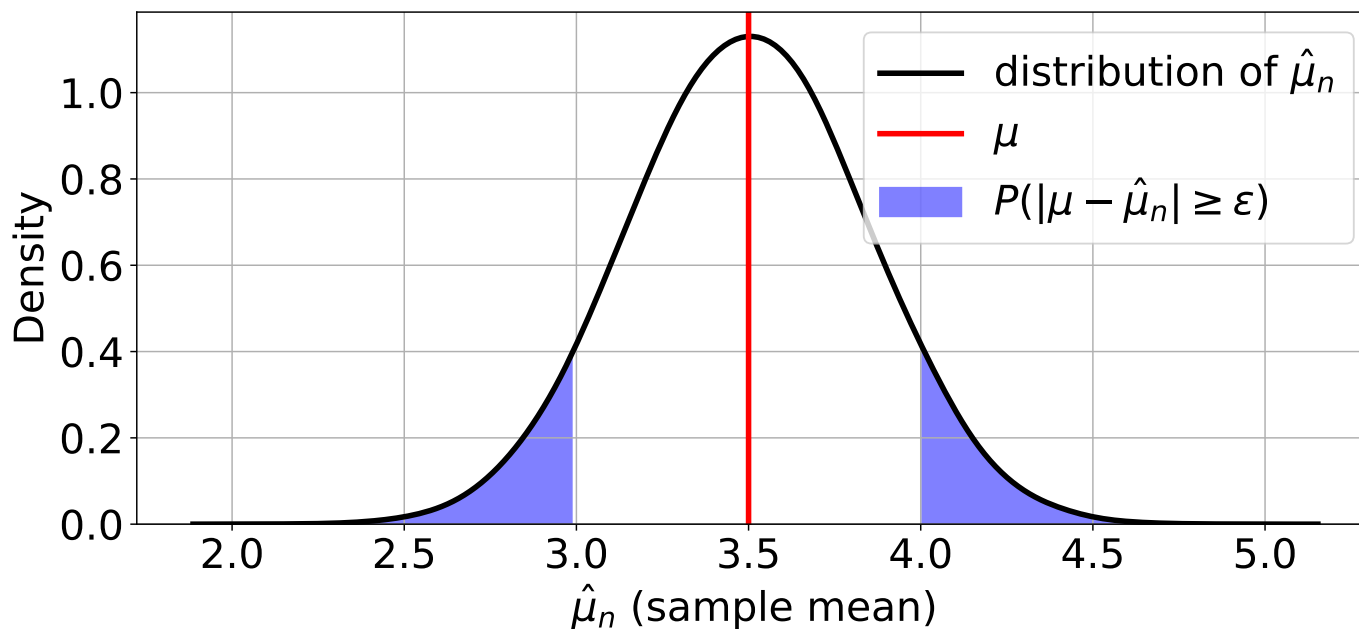
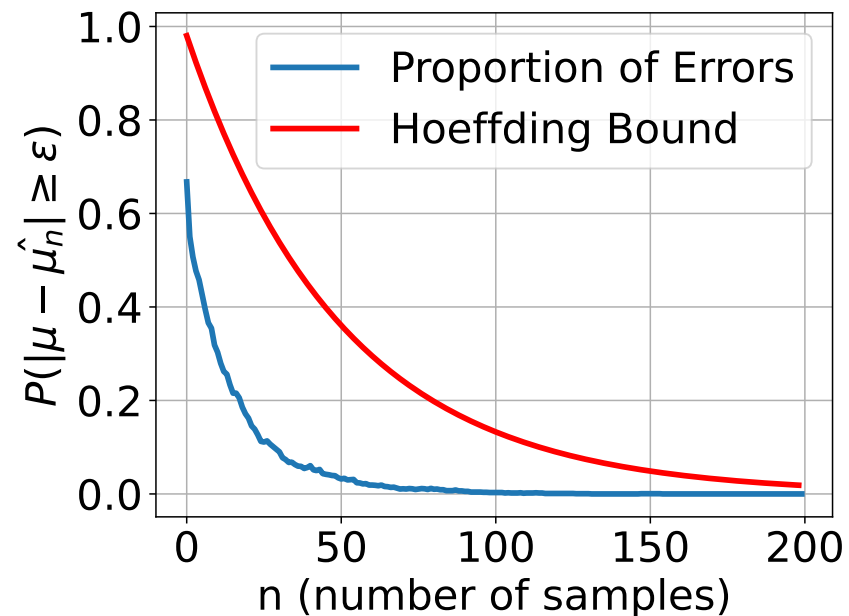
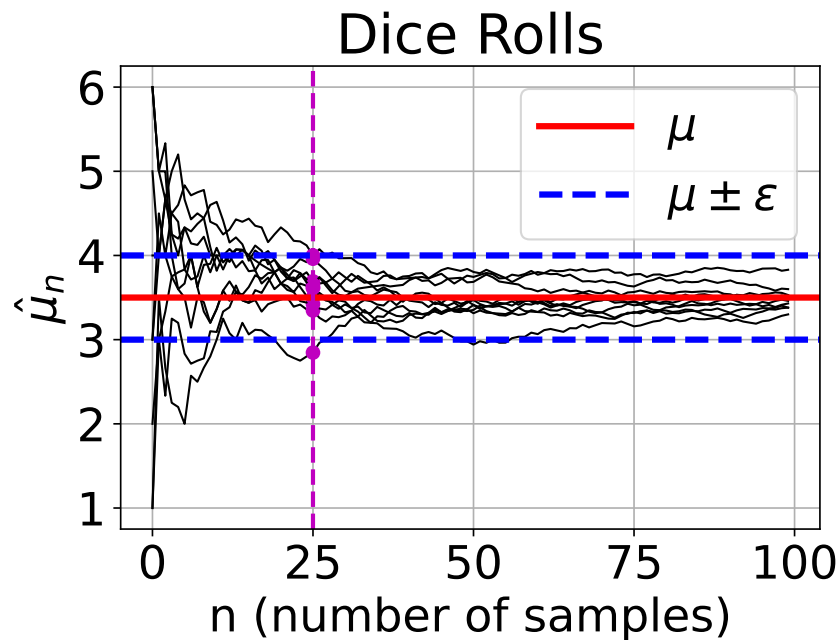
$$\mathbb{E}_{Z_n \sim q^n}[\hat{\mu}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n z_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[z_i] = \mu$$

Sample mean as an estimator of the expected value

- ◆ Let $Z_n = (z_1, \dots, z_n)$ be i.i.d. samples drawn from rand. vars. with the expected value $\mu = \mathbb{E}_{z \sim q}[z]$ and variance $\sigma^2 = \mathbb{V}_{z \sim q}[z]$.
- ◆ The variance $\mathbb{V}_{Z_n \sim q^n}[\hat{\mu}_n]$ of the sample mean $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n z_i$ decays with $\frac{1}{n}$:

$$\begin{aligned}
\mathbb{V}_{Z_n \sim q^n}[\hat{\mu}_n] &= \mathbb{E} \left[\left(\frac{1}{n} \sum_{i=1}^n z_i - \mu \right)^2 \right] \\
&= \mathbb{E} \left[\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (z_i - \mu)(z_j - \mu) \right] \\
&= \frac{1}{n^2} \mathbb{E} \left[\sum_{i=1}^n (z_i - \mu)^2 \right] + \frac{1}{n^2} \sum_{i \neq j} \underbrace{\mathbb{E}[(z_i - \mu)(z_j - \mu)]}_{=0 \text{ because } z_i \text{ and } z_j \text{ are independent}} \\
&= \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}(z_i) \\
&= \frac{\sigma^2}{n}
\end{aligned}$$

Hoeffding inequality



Hoeffding inequality:

$$\mathbb{P}(|\mu - \hat{\mu}_n| \geq \varepsilon) \leq 2e^{-\frac{2n\varepsilon^2}{(b-a)^2}}$$

$$a = 1, b = 6, \varepsilon = 0.5$$

Hoeffding inequality

- ◆ **Theorem:** Let $Z_n = (z_1, \dots, z_n)$ be i.i.d. samples drawn from r.v. with distribution $q(z)$. Let the random variables attain values from an interval $[a, b]$. Let the expected value of the r.v. be $\mu = \mathbb{E}_{z \sim q}[z]$. Let $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n z_i$ be the sample average. Then, for any $\varepsilon > 0$:

$$\mathbb{P}_{Z_n \sim q^n} \left(|\hat{\mu}_n - \mu| \geq \varepsilon \right) \leq 2e^{-\frac{2n\varepsilon^2}{(b-a)^2}}$$

◆ **Key Properties:**

- (+) **General:** the bound holds for any bounded i.i.d. random variables.
- (-) **Conservative:** the bound is typically not tight.
- (+) **Vanishing:** the bound $\rightarrow 0$ as $n \rightarrow \infty$.
- (+) **Cheap:** the bound is simple and easy to compute.

Weak Law of Large Numbers

Theorem: Let $Z_n = (z_1, \dots, z_n)$ be i.i.d. sample generated from r.v. with a distribution $q(z)$. Let the random variables attain values from an interval $[a, b]$. Let the expected value of the r.v. be $\mu = E_{z \sim q}[z]$. Let $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n z_i$ be the sample average. Then, for any $\varepsilon > 0$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\hat{\mu}_n - \mu| > \varepsilon) = 0$$

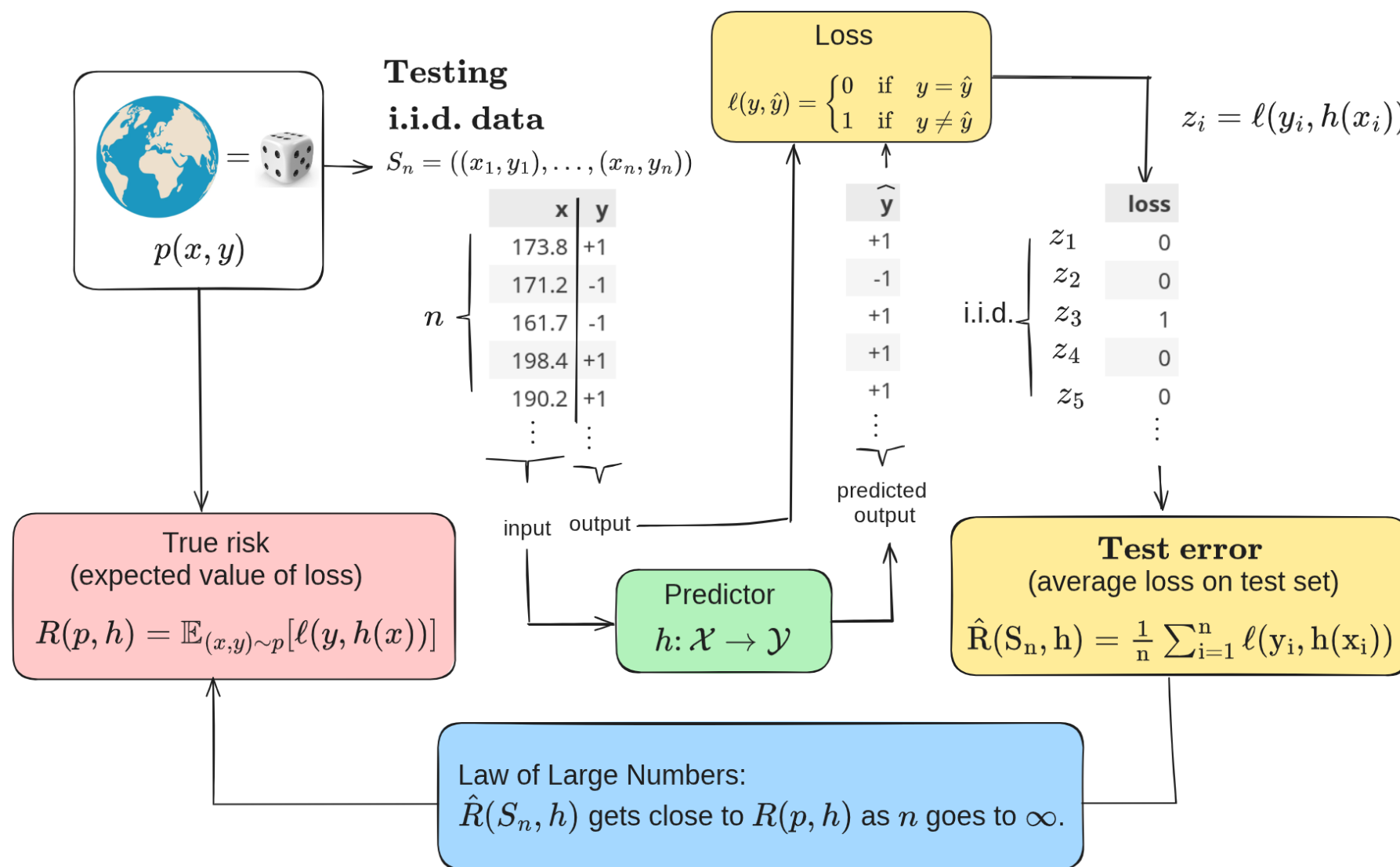
i.e., the sequence $\{\hat{\mu}_n\}_{n=1}^{\infty}$ **converges in probability** to μ , which is denoted by

$$\hat{\mu}_n \xrightarrow{P} \mu .$$

Interpretation:

- ◆ As the number of samples increases, the sample average becomes arbitrarily close to the expected value with high probability.
- ◆ The probability of large deviations from μ vanishes as $n \rightarrow \infty$.
- ◆ Empirical averages approximate population expectations.

Predictor evaluation



Application of the Hoeffding inequality:

$$\mathbb{P}_{Z_n \sim q^n} \left(|\hat{\mu}_n - \mu| \geq \varepsilon \right) \leq 2e^{-\frac{2n\varepsilon^2}{(b-a)^2}} \Rightarrow \mathbb{P}_{S_n \sim p^n} \left(|\hat{R}(S_n, h) - R(p, h)| \geq \varepsilon \right) \leq 2e^{-\frac{2n\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}}$$

Confidence interval

- ◆ **Goal:** Characterize the deviation between the true risk $R(p, h) = \mathbb{E}_{(x,y) \sim p}[\ell(y, h(x))]$ and the test error $\hat{R}(S_n, h) = \frac{1}{n} \sum_{i=1}^n \ell(y^i, h(x^i))$ computed on i.i.d. data $S_n \sim p^n$.
- ◆ **Hoeffding inequality:** provides a probabilistic bound on the deviation between the true risk $R(p, h)$ and the test error $\hat{R}(S_n, h)$:

$$\mathbb{P}_{S_n \sim p^n} \left(|R(p, h) - \hat{R}(S_n, h)| \geq \varepsilon \right) \leq 2e^{-\frac{2n\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}} \quad \forall \varepsilon > 0$$

- ◆ **(1- δ)-Confidence interval:** (derived from the Hoeffding inequality)

$R(p, h) \in (\hat{R}(S_n, h) - \varepsilon, \hat{R}(S_n, h) + \varepsilon)$ holds with probability $1 - \delta$ at least,

where $1 - \delta$ is called the confidence level.

- For fixed n and $\delta \in [0, 1]$, compute $\varepsilon = (\ell_{\max} - \ell_{\min}) \sqrt{\frac{\log(2) - \log(\delta)}{2n}}$
- For fixed ε and $\delta \in [0, 1]$, compute $n = \frac{\log(2) - \log(\delta)}{2\varepsilon^2} (\ell_{\max} - \ell_{\min})^2$

Summary

◆ Law of Large Numbers

- The sample mean converges to the expected value as the number of samples increases.
- The sample mean is an unbiased estimator of the expected value.
- The variance of the sample mean decreases at the rate $\frac{1}{n}$:
- Hoeffding's inequality provides bounds the probability of deviation between the sample mean and the expected value.
- The Weak Law of Large Numbers (convergence in probability).

◆ Predictor Evaluation

- For a fixed $h: \mathcal{X} \rightarrow \mathcal{Y}$, the losses $\ell(y_i, h(x_i))$ on an i.i.d. sample S_n are themselves i.i.d. random variables.
- The sample mean of the losses is the *empirical risk*; the expected value is the *true risk*.
- Confidence intervals provide probabilistic bounds on the error of estimation.